

SANTOPRENE® 103-40

A hard, black, versatile thermoplastic vulcanizate (TPV) in the thermoplastic elastomer (TPE) family. This material combines good physical properties and chemical resistance for use in a wide range of applications. This grade of Santoprene® TPV is shear-dependent and can be processed on conventional thermoplastics equipment for injection molding, extrusion, blow molding, thermoforming or vacuum forming. It is polyolefin based and recyclable within the manufacturing stream.

Key Features

- UL listed: file #QMFZ2.E80017, Plastics - Component; file #QMFZ8.E80017, Plastics Certified For Canada - Component
- Excellent ozone resistance

Typical mechanical properties

Stress at 100% elongation, perpendicular	9 MPa	ISO 527-1/-2 or ISO 37
Stress at break, perpendicular	20.7 MPa	ISO 527-1/-2 or ISO 37
Elongation at break, perpendicular	610 %	ISO 527-1/-2 or ISO 37
Brittleness Temperature	-52 °C	ASTM D 746
Low temperature brittleness	-52 °C	ISO 812
Shore D hardness, 15s	41	ISO 48-4 / ISO 868
Shore D hardness, 168h at 150°C	4	ISO 48-4 / ISO 868
Compression set at 70°C, 24h	54 %	ISO 815
Compression Set at 125°C, 70h	61 %	ISO 815
Tear strength, normal	64 kN/m	ISO 34-1

Thermal properties

RTI, electrical, 1.5mm	85 °C	UL 746B
RTI, electrical, 3.0mm	85 °C	UL 746B
RTI, strength, 1.5mm	85 °C	UL 746B
RTI, strength, 3.0mm	85 °C	UL 746B

Specific Application Suitability

Detergent resistance	f3	UL 749
Detergent resistance	f4	UL 2157

Flammability

Burning Behav. at 1.5mm nom. thickn.	HB class	UL 94
Thickness tested	1.5 mm	UL 94
UL recognition	yes	UL 94
Burning Behav. at thickness h	HB class	UL 94
Thickness tested	1.00 mm	UL 94
UL recognition	yes	UL 94
Hot Wire Ignition, 1mm	PLC 4 s	UL 746A
Hot Wire Ignition, 1.5mm	PLC 3 s	UL 746A
Hot Wire Ignition, 3mm	PLC 2 s	UL 746A

SANTOPRENE® 103-40

Electrical properties

Relative permittivity, 60Hz	2.6	IEC 62631-2-1
Comparative tracking index, 23 °C	PLC 0 PLC	UL 746A
Arc Resistance Performance Level Category	PLC 5 class	UL 746B
Electric Strength, Short Time, 2mm	32 kV/mm	ASTM D 149
High Amperage Arc Ignition Category, 1.5 mm	PLC 0 class	UL 746A
High Voltage Arc Tracking Rate	PLC 1 mm/min	UL 746A

Physical/Other properties

Density	950 kg/m³	ISO 1183
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Injection

Drying Temperature	82 °C	
Drying Time, Dehumidified Dryer	3 h	
Processing Moisture Content	0.08 %	
Max. regrind level	20 %	
Melt Temperature Optimum	215 °C	Internal
Max. mould temperature	10 - 52 °C	
Vent depth	25 µm	
Back pressure	0.345 - 0.689 MPa	
Injection speed	fast	

Extrusion

Drying Temperature	82 °C
Drying Time, Dehumidified Dryer	3 h
Melt Temperature Range	210 °C

Processing Texts

Processing Notes	Desiccant drying for 3 hours at 80 °C (180 °F) is recommended. Santoprene® TPV has a wide temperature processing window from 175 to 230 °C (350 to 450 °F) and is incompatible with acetal and PVC.
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Other Approvals

Other Approvals	OEM	Specification	Additional Information
	Stellantis - Chrysler	MS-AR-100 FGN	
	Ford	WSD-M2D441-A	
	GM	GMW15813, Type 9	
	Stellantis - PSA Group	PMP 01994_10_00138	
	Mercedes-Benz Group (Daimler)	DBL 5562	

SANTOPRENE® 103-40

Stellantis - FCA Group	55248/02	EMP-140
Stellantis - FCA Group	B63 0300 01378_23_00032	Coolant hose celanese TPV1hose-125
Hyundai	MS220-05, Type G	
Renault	FRM 18-27-134 /---	
Renault	UB02b	PMR2021
Renault	UM09g	PMR2021
VW Group	VW50123	
ZF Group	ZF 7832.050.153	

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